

REPORT 1

Project Decisions & Architecture

Intelligent Graph Analytics Platform

Project Report - Part 1

Methods & Design Decisions

1. Problem Statement

Organizations frequently receive large CSV datasets containing business information such as customer records, sales transactions, healthcare data, financial reports, and logistics information. Extracting meaningful insights from these datasets requires technical expertise in SQL or business intelligence tools, making data exploration difficult for non-technical users.

Our objective is to build an intelligent graph-based analytics platform that automatically converts uploaded CSV datasets into a graph database and enables users to query the data using natural language.

2. Objectives

Objective Description

CSV Upload Accept any structured CSV dataset

Data Streaming Process uploaded data using Apache Kafka

Graph Construction Convert CSV rows into Neo4j graph nodes

Conversational Analytics Allow users to ask questions in natural language

Dashboard Generation Automatically summarize uploaded datasets

Scalable Deployment Deploy using Docker containers

3. Solution Overview

Our solution is divided into five independent services.

Module Responsibility

Frontend CSV Upload, Dashboard, Chat Interface

FastAPI CSV Validation, API Management

Kafka Real-time Message Streaming

Loader Convert CSV into Graph

Neo4j Store Graph Database

4. Why We Chose This Architecture

Decision Reason

Microservices Independent deployment and scalability

Docker Easy setup and reproducibility

Kafka Reliable asynchronous processing

Neo4j Best suited for relationship-based querying

FastAPI High-performance REST APIs

Next.js Modern responsive frontend

5. Alternatives Considered

Component Selected Alternative Reason

Database Neo4j MySQL Graph relationships are easier to query

Queue Kafka RabbitMQ Better streaming support

Backend FastAPI Flask Faster and asynchronous

Frontend Next.js React Routing and optimization

Deployment Docker Manual Installation Easy portability

6. Overall Workflow

User Uploads CSV

?

FastAPI

?

CSV Validation

?

Kafka

?

Loader Service

?

Neo4j Graph

?

Graph Ready

?

Chatbot

?

Cypher Query

?

Answer

?

Dashboard

7. System Architecture

Next.js Frontend

Upload CSV Chat Interface

?

?

FastAPI API

?

CSV Validation

?

?

Apache Kafka

?

Message Streaming

?

Loader Service

?

Graph Construction

?

Neo4j

?

Graph Query Engine

?

Dashboard + Chatbot

8. Design Principles

? Modular

? Scalable

? Fault Tolerant

? Dataset Independent

? Containerized

? Easy Deployment

? Fast Response

9. Technology Stack

Layer Technology

Frontend Next.js 15

Backend FastAPI

Streaming Apache Kafka

Database Neo4j

Language Python

UI Tailwind CSS

Deployment Docker

API REST

10. Data Flow

CSV

?

Rows

?

Kafka Messages

?

Loader

?

Neo4j Nodes

?

Relationships

?

Graph

?

Cypher

?

Results

11. Key Features

Automatic CSV ingestion

Real-time Kafka streaming

Dynamic graph generation

Neo4j knowledge graph

Natural language chatbot

Dynamic dataset switching

Dashboard generation

Report generation

Docker deployment

REST APIs

12. Innovation

Unlike traditional CSV analytics tools, our platform transforms structured tabular data into a graph database and provides conversational access through an intelligent chatbot. The system dynamically adapts to newly uploaded datasets without requiring predefined schemas, enabling users to explore relationships and insights using natural language. The integration of Apache Kafka, Neo4j, FastAPI, and Docker delivers a scalable, modular, and production-ready architecture suitable for enterprise analytics.

13. Expected Output

After uploading a dataset, users can:

View dataset statistics

Explore column information

Ask questions in natural language

Retrieve graph-based insights

Generate automatic dashboards

Visualize key metrics

Query multiple uploaded datasets

14. Conclusion

The proposed architecture provides a modern, scalable, and intelligent approach to CSV analytics. By combining streaming, graph databases, and conversational interfaces, the platform simplifies data exploration while maintaining modularity, performance, and ease of deployment.